

TEST - 3
GENERAL STUDIES - 1
ASCEND - GS MAINS TEST SERIES 2024

Q1: Poverty encompasses not only financial deprivation but also social marginalization and exclusion.
Comment (10 Marks, 150 Words)

Approach to the answer:

Introduction: Introduce by defining poverty.

Body: Address the demand of the question by how lack of financial resources contributes to poverty and then mention how other factors i.e. social marginalization and exclusion contribute to poverty.

Conclusion: Conclude with a summarizing the topic on positive note.

Answer: The UN defines poverty as a human condition characterized by sustained or chronic deprivation of resources, capabilities, choices, security, and power necessary for enjoying an adequate standard of living. Poverty is a multidimensional problem that includes social marginalization and exclusion as well.

Lack of financial resources contributes to poverty:

- **Access to Basic Necessities:** People in poverty often lack the means to access essentials such as food, clean water, and shelter, leading to poor health outcomes and perpetuating the cycle of poverty.
- **Quality Education:** Insufficient financial resources hinder access to quality education, limiting opportunities for employment thereby reinforcing intergenerational poverty.
- **Healthcare:** Financial constraints prevent access to healthcare services, resulting in increased disease burden, reduced productivity, and lower incomes.
- **Financial Exclusion:** Lack of financial resources leads to indebtedness to loan sharks, reduces savings, and limits opportunities for entrepreneurship.

Social marginalization and exclusion contribute to poverty:

- **Limited Access to Education:** Marginalized groups often lack quality education, hindering economic opportunities.
 - For Example: Dalits in rural areas face educational barriers due to caste discrimination.
- **Social Stigma:** Scheduled Castes and Scheduled Tribes experience limited social mobility, access to credit, and job opportunities due to caste-based exclusion.
- **Discrimination Against Minorities:** Discrimination based on factors like caste, religion and gender restricts opportunities and perpetuates poverty.
- **Political Marginalization:** Limited political representation hinders effective policy responses to poverty, exacerbating marginalization and persecution of marginalized groups.

Addressing poverty requires a holistic approach that goes beyond income generation to tackle social marginalization and exclusion. Only by addressing these multidimensional aspects can meaningfully progress towards sustainable development and equality be achieved.

Q2: Analyse the ways in which urbanization has impacted social values and traditions. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with definition of urbanization and its impact.

Body: Address the demand by listing positive and negative impacts of urbanization on values and traditions. Followed by ways in which urban planning that can promote social inclusion and cultural diversity.

Conclusion: Conclude with summarizing the topic.

Answer: Urbanization refers to the growth in the number of people living in cities as opposed to villages. Urbanization has reshaped social values and traditions by bringing about both positive advancements and challenges as societies transition from rural to urban lifestyles.

Positive Impacts on Values and Traditions

- **Diversity and Pluralism:** Exposure to diverse cultures fosters tolerance, acceptance, and appreciation for diversity.
- **Modern values:** Urbanization promotes modern values like equality and individualism.
- **Social Reforms:** Economic constraints diminish the influence of caste norms, fostering anonymity and social mobility.
- **Access to Amenities:** Urbanization improves access to education, healthcare, and job opportunities.
- **Changing Gender Roles:** Urbanization tends to empower women economically and socially. It challenges traditional gender roles and promotes equality in decision-making.

Negative Impacts on Values and Traditions

- **Social Fragmentation:** Rapid urbanization can lead to social fragmentation and alienation, especially among the elderly.
- **Individualism:** Individualism in urban settings can alter familial dynamics, community cohesion, and societal responsibility.
- **Materialism:** Materialistic values placing greater importance on material possessions in cities that lead to increased crime rates and social unrest.
- **Environmental Degradation:** Urban expansion often results in environmental degradation through consumerism, pollution, and habitat destruction.
- **Impact on Traditional Practices:** Urban migration often leads to the erosion of traditional practices and customs as younger generations may distancing themselves from rural traditions.

Urbanization must foster inclusive urban planning, promoting cultural preservation alongside progress, and strengthening community bonds which is essential for navigating the complexities of urbanized societies.

Q3: The Indian model of secularism has successfully fostered an inclusive and integrated society. Examine (10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with defining communalism or else secularism.

Body: Address the demand by highlighting – the role of Indian model of secularism in to building an inclusive and integrated society. Lastly, suggest ways to make secularism in India more resilient.

Conclusion: Conclude with summarizing the topic.

Answer: Indian secularism treats all religions equally and permits the state to interact with religions within legal boundaries. It does not completely separate religion from state affairs, allowing for some state involvement in religious matters as long as it stays within the law.

Indian model of secularism creating an Inclusive Society:

- **Equal Treatment of All Religions:** Indian secularism ensures equal respect and treatment for all religions.
- **Protection of Minority Rights:** Special provisions exist to protect the cultural and educational rights of religious minorities.
 - Example: Article 30 allows minorities to establish and administer educational institutions.
- **Secular Legal Framework:** Laws are designed to be secular, not favoring any particular religion.
 - Example: The Special Marriage Act allows interfaith marriages without religious conversion.
- **Protects vulnerable groups:** It protects vulnerable groups like women while balancing with religious rights or personal laws, for example by the abolition of practices like triple talaq.
- **Constitutional Morality:** It upholds constitutional morality by ensuring equality, as seen in decisions like allowing women's entry into Sabarimala temple case.

Challenges:

- **Inconsistent Application:** Secularism is applied inconsistently, leading to perceived biases.
 - Example: Different personal laws for various religious communities.
- **Political Manipulation:** Politicians often use religion for electoral gains, undermining secular principles.
 - Example: Vote bank politics targeting specific religious communities.
- **Judicial Overreach:** Courts sometimes struggle to balance religious freedom with constitutional values.

- **State interventions:** in religious matters lead to feelings of alienation among certain communities.

To strengthen the Indian model of secularism, it is crucial to codify and reform personal laws to align with Fundamental Rights. State initiatives like "Jiyo Parsi," "Nai Roshni," and "Ek Bharat, Shrestha Bharat" actively promote minority welfare and prevent alienation for sustaining a harmonious and integrated society.

Q4. Critically examine the arguments both in favour of and against legalizing same-sex marriage in India. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Introduce by giving the recent context of same-sex marriages in India (SC case).

Body: Mention how same-sex marriage is seen as against Indian culture and society. Give arguments in favor of legalizing same-sex marriages in India.

Conclusion: Conclude with a futuristic viewpoint showing evolution of society on progressive lines.

Answer: Same sex marriage refers to the legal and social recognition of marriage between individuals of the same gender or sexual orientation. In India, same-sex marriage is not legally recognized.

Arguments in favour of same sex marriage:

- **Protection of Rights:** The Delhi High Court's Naz Foundation vs. Govt. of NCT of Delhi (2009) ruling declared Section 377 unconstitutional, protecting LGBTQ+ rights.
- **Personal Freedom:** Individuals have the right to marry based on their sexual orientation.
 - In 2017, the Supreme Court's K.S. Puttaswamy judgment affirmed that sexual orientation is a crucial part of one's identity and must be protected from discrimination.
- **Legal Precedents:** In the Navtej Singh Johar case, the Supreme Court affirmed that LGBT individuals are entitled to full constitutional rights.
- **Strengthening Family Bonds:** Legal marriage strengthens family bonds and provides legal protections.
 - Example: Legal recognition can facilitate access to healthcare benefits and inheritance rights.

Arguments against same-sex marriage:

- **Religious Freedoms:** Legalizing same-sex marriage may infringe on religious freedoms and practices.
 - Example: Legal challenges from religious institutions citing freedom to define marriage.
- **Cultural Concerns:** Opponents of same-sex marriage argue that it goes against customs and traditions. Some view same-sex marriage as an urban elitist idea.
- **Legal Complexity:** Redefining marriage requires complex legal changes and implications.
- **Adoption Challenges:** The Supreme Court upheld existing laws, denying adoption rights to same-sex couples.

Same-sex marriages debate represent a growing push towards inclusivity and equality. There is need for evolving dynamics between cultural traditions, legal frameworks, and changing societal attitudes.

Additional Information:

Several high court verdicts have ruled in favor of same sex couples:

- **In Madhu Bala vs State of Uttarakhand (2020)**, the high court of Uttarakhand held that right of a same sex couple to live together is a constitutional and human right.
- **In Vanitaben Damjibhai Solanki vs State of Gujarat (2020)**, the Gujarat High Court ordered police protection for two women police constables in a relationship.
- **In S Sushma v Commissioner of Police (2021)**, the court protects the couple in relationship and makes sure that both sets of parents are taken along in this journey.

Supreme Court Hearing: Views on Four Key Issues

- **Adoption Rights for Queer Couples:** The Supreme Court upheld existing laws denying adoption rights to same-sex couples in a split 3:2 verdict.
- **Fundamental Right to Marriage:** The Supreme Court unanimously ruled that marriage is not an

absolute right and cannot be claimed as fundamental right by same-sex couples.

- **Interpretation of Special Marriage Act:** The Supreme Court unanimously rejected challenges to provisions of the Special Marriage Act regarding same-sex marriage.
- **Option of Civil Union for Queer Couples:** The majority opinion supported enabling civil unions for same-sex couples, while minority opinions by CJI Chandrachud and Justice Kaul advocated for recognizing these relationships as civil unions with associated benefits.

Q5: Can India's demographic dividend potentially turn into a demographic burden in the future? Justify your answer (10 Marks, 150 Words)

Approach to the answer:

Introduction: Introduce by defining the demographic dividend.

Body: Address the demand of the question by highlighting how India's demographic dividend could turn into a demographic burden then mention what leads to harnessing demographic dividend.

Conclusion: Conclude with a summarizing the topic on positive note.

Answer: The demographic dividend refers to the economic growth potential that arises from shifts in a population's age structure. This occurs when the proportion of the working-age population (15 to 64) exceeds that of the non-working-age population (14 and younger, and 65 and older). Around 68% of India's population falls between the ages of 15 and 64.

India's Demographic Dividend could turn into a Demographic Burden:

- **Poor Skilling:** Despite efforts, skilling initiatives have not kept up with population growth, resulting in a significant skills gap.
 - For example, only one in five Indians in the labor force is considered "skilled" as per Human Development Report 2022.
- **Unemployment:** Even educated and skilled youth struggle to find employment.
 - For example, Unemployment rate in India rose sharply to 9.2 per cent in June 2024
- **Rising Dependency Ratio:** India's old-age dependency ratio is projected to increase from the current 19% to around 35% by 2050.
- **Shift from Primary Sector:** Realizing the demographic dividend requires shifting workers from agriculture to secondary and tertiary sectors.
 - The farm sector's share of the employed labor force was 45.5% in 2021-22, as per Periodic Labour Force Survey.
- **Health Issues:** Poor health limits educational and professional prospects.
 - For example, over one-third of children (35.5%) are stunted as per NFHS-5.

Harnessing India's Demographic Dividend:

- According to a CII report, leveraging this demographic advantage could increase India's GDP from the current \$3 trillion to \$9 trillion by 2030.
- However, achieving these economic milestones depends on efforts in skilling and reskilling the workforce.
- Enhancing job creation across various sectors is crucial to absorb the growing working-age population.

India's demographic dividend presents significant economic opportunities, realizing its full potential requires proactive policies in healthcare, education, skill development, and job creation for transforming the demographic dividend.

Q6. Highlight the global distribution of volcanoes and explain why they are primarily concentrated along the Pacific Ring of Fire. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Give a brief definition of volcano.

Body: Address the demand highlighting Global distribution of Volcanoes and mention reason about their concentration along Pacific Ring of Fire.

Conclusion: Summarizes with topic of volcano.

Answer: A volcano is an opening in the Earth's crust where gases, ash, and molten lava escape from a magma chamber below the surface. The process of a volcano erupting is called volcanism.

- Volcanoes are found mostly in areas where the Earth's crust is weaker and prone to seismic activities like earthquakes.

Global Distribution of Volcanoes:

- **Circum-Pacific Belt (Ring of Fire):** There are volcanoes along the eastern and western coasts of the Pacific Ocean.
 - **Examples:** Fujiyama in Japan, Mount St. Helens in the USA, etc.
 - Around 75% of active volcanoes worldwide are found along the Ring of Fire.
- **Mid-Continental Belt:** These volcanoes are found in the Alpine mountains and around the Mediterranean Sea. They occur due to the collision of the Eurasian plate with the African and Indian plates.
 - **Examples:** Stromboli, Vesuvius, and Etna volcanic mountains.
- **Mid-Atlantic Belt:** This includes volcanoes along the mid-Atlantic ridge, a divergent plate zone. They mostly have fissure eruptions.
 - **Examples:** Iceland is the most active volcanic area in this belt.



The high number of volcanoes along the Ring of Fire is due to:

- **Presence of many tectonic plates:** The area has several tectonic plates, including the Pacific, Philippine, Cocos, Nazca, and North American plates.
 - The movement of these plates leads to frequent earthquakes and create conditions conducive to volcanic activity.
- **Subduction zones:** Along the Ring of Fire, where one tectonic plate is forced beneath another called subduction zones.
 - As the subducting plate descends, it melts and forms magma, which rises to the surface to create volcanoes.
- **Divergent Boundaries and Hotspots:** The Ring of Fire also includes areas where tectonic plates are pulling apart (divergent boundaries) and hotspots where plumes of hot mantle material rise.

Volcanoes are globally distributed along tectonic plate boundaries due to the presence and movement of multiple tectonic plates and subduction zones, which create ideal conditions for frequent volcanic activity and earthquakes.

Q7: Highlight the factors that contribute to the shifting of wind belts and discuss how these shifts impact the climate of a region. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Introduce in brief about mechanism that leads to shifting of wind and pressure belt.

Body: Address the demand highlighting shift of pressure and wind belt and then mention climatic conditions changed by seasonal reversal of wind.

Conclusion: Conclude with summarizes with topic.

Answer: The Earth's axial tilt and its revolution around the sun are key factors in the shifting of wind belts. As the Earth orbits the sun, different parts of the planet receive varying amounts of sunlight throughout the year, causing changes in temperature and pressure patterns leading to the shifting of wind belts.

Factors contribute to the shifting of wind belts:

- **Seasonal Temperature Variations:** Differential heating of land and sea alters pressure systems.
 - Example: Summer and winter solstices shift pressure belts and wind patterns globally.
- **Global Pressure Systems:** Changes in high and low-pressure areas drive wind belt movements.
 - Example: El Niño events affecting pressure patterns and wind directions across continents.
- **Oceanic Circulation:** Ocean currents influence wind directions along coastlines and islands.
 - Example: The Gulf Stream impacting westerlies in the North Atlantic Ocean.
- **Climate Oscillations:** Long-term cycles like the Arctic Oscillation impact wind belt shifts.
 - Example: Changes in the North Atlantic Oscillation influencing jet streams and weather patterns.

Climatic Conditions changed by seasonal reversal in Wind Belts:

- **Precipitation Patterns:** Shifts in wind belts alter rainfall distribution.
 - Example: South-western winds bring heavy summer rains in India.
- **Temperature Extremes:** Altered wind belts lead to temperature anomalies.
 - Example: The positive Indian Ocean Dipole led to hotter and drier conditions in Australia.
- **Weather Events:** Variations in wind belts contribute to extreme weather phenomena.
 - Example: Changes in jet streams causing prolonged droughts or heavy rainfall.
- **Ecological Effects:** Changes in wind belts impact species distribution.
 - Example: Changes in the Humboldt Current off the coast of Peru affect marine life, influencing fisheries and seabird populations.

The shifting of wind belts due to the Earth's axial tilt and its revolution around the sun plays a crucial role in determining regional climates. Understanding these mechanisms helps in predicting climatic changes and preparing for their impacts on various regions.

Q8: Explain why Himalayan rivers form large deltas as compared to peninsular rivers. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Introduce in brief about river delta formation.

Body: Address the demand highlighting factors which led to Himalayan rivers form larger deltas compared to peninsular rivers.

Conclusion: Conclude with summarizes with topic.

Answer: A river delta forms at the mouth of a river where it meets an ocean or larger water body, characterized by low-lying plains. Deltas are vital for human activities, fisheries, and wildlife due to their fertile soil and dense vegetation.

Himalayan rivers form larger deltas compared to peninsular rivers due to several factors:

Aspect	Himalayan Rivers	Peninsular Rivers
Size and Speed	Larger, faster flow carrying substantial sediment loads. Example: Ganga and Brahmaputra rivers forming Sundarbans delta.	Smaller, slower flow with lesser sediment transport capacity.
Sediment Load	Fed by numerous tributaries with significant sediment accumulation. Example: Ganga receives sediment from tributaries such as Yamuna, Ghaghara, and Gandak.	Receive sediment from local terrain with limited tributary contributions.
Erosion	Significant erosion due to softer terrain and higher flow velocities.	Slower erosion over harder rock surfaces, less sediment contribution.
Gradient	Steep gradients accelerate downstream transport of sediment.	Gentle gradients reduce sediment transport efficiency.
Perennial Flow	Sustained flow throughout the year due to monsoon rains and snowmelt.	Seasonal flow patterns with reduced water and sediment transport in dry seasons.

This comparison underscores why Himalayan rivers, such as the Ganga and Brahmaputra, can form larger and more extensive deltas compared to peninsular rivers like the Godavari or Krishna.

Q9: India's wind energy potential remains largely unexploited due to various reasons. Discuss and suggest a way forward. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Introduce in brief about wind energy and relevant data.

Body: Address the demand highlighting factors which contribute to underutilization of wind potential and then suggest way forward.

Conclusion: Conclude with summarizes with topic.

Answer: India has a lot of wind energy potential, estimated at 302 GW at 100 meters height and expanding to 695 GW at 120 meters. There is also offshore wind energy potential of 174 GW.

However, due to slow wind energy installations about 13.5 GW renewable energy capacity added during calendar year 2023, bringing the total installed capacity to 43.7 GW, which is much lower than what could be achieved.

Factors contribute to this underutilization:

- **Infrastructure Challenges:** Lack of grid connectivity in remote windy areas hampers development.
 - Example: Wind-rich regions like Ladakh lack sufficient transmission infrastructure.
- **Gap between wind and solar energy:** Declining costs of solar power panels have widened the economic disparity between solar and wind energy.
- **Delays in land acquisition:** The installation of onshore wind turbines necessitates large plots of land, often encountering opposition.
 - Example: Delays in land acquisition have stalled wind projects in coastal Gujarat.
- **Financial Viability:** Offshore wind turbines are estimated to cost two to three times more per megawatt than onshore turbines, posing substantial financial challenges.
- **Policy Uncertainty:** Changing regulations and inconsistent incentives deter investment.
 - Example: Uncertainty over wind tariff policies affects investor confidence.

Addressing these challenges requires coordinated efforts:

- **Centre-state coordination:** Upgrade grid infrastructure, stable tariff policies, long term power purchase agreement and establish clear conditions for new tenders.
- **Overcoming legacy issues:** Address challenges like delayed payments, inadequate grid infrastructure, evolving state land acquisition policies.
- **Promote Research:** Fund research for more efficient wind turbine technologies.
- **Offshore wind power:** Develop comprehensive offshore wind roadmaps, particularly in Gujarat and Tamil Nadu, to harness their substantial potential.

India's commitment to achieve net-zero emissions by 2070 and increase non-fossil fuel energy capacity to 50% by 2030 underscores the critical importance of investing in renewable energy sources, particularly wind power.

Q10: Discuss the factors contributing to the significant concentration of the pharmaceutical industry on India's western coast. (10 marks, 150 words)

Approach to the answer:

Introduction: Write the nature of pharmaceutical industry.

Body: Factors Responsible for regional concentration of pharmaceutical industry.

Conclusion: Provide new trends in the location of pharmaceutical industries

Answer: The pharmaceutical industry is typically flexible and doesn't rely on specific local resources that are heavy or light in weight. Despite this, most of India's pharmaceutical activity is concentrated in Maharashtra and Gujarat.

Factors contributing for this concentration:

- **Proximity to Ports:** Easy access to import raw materials and export finished products to markets like Africa and Europe through ports such as Kandla, Mumbai, and Bhavnagar.
- **Nearness to Petrochemical Hubs:** Access to essential petrochemicals needed in pharmaceutical production, found near the Gujarat Coast and Mumbai.
- **Historical Factors:** Established pharmaceutical clusters attract new investments and collaborations. Example: Mumbai and Ahmedabad
- **Skilled Workforce:** Presence of top pharmaceutical training institutes in Maharashtra, like Bombay College of Pharmacy and Government College of Pharmacy, attracts talented professionals.
- **Infrastructure:** Well-developed infrastructure including roads, railways, airports, ports, and Special Economic Zones (SEZs) on the western coast.
- **Supportive Government Policies:** States like Gujarat and Maharashtra offer business-friendly policies, tax breaks, and quick approvals for setting up pharmaceutical plants.
- **Access to Capital:** Western states like Maharashtra, known for Mumbai as the financial hub, offer good banking systems that support long-term funding for research and development.



Recently, there has been some industry diversification to hilly regions due to favorable climate conditions and to Hyderabad due to supportive state policies. The Indian government has also introduced Production Linked Incentives to promote pharmaceutical industry growth nationwide.

Q11: How do cultural and social norms influence population growth in India? Assess the necessity of a population control law in India. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with data on India's population or with the definition of population growth.

Body: Address the demand by highlighting – various Social and cultural norms influence population growth followed by – do we have a necessity of population control law.

Conclusion: Conclude with way forward to address stressed migration.

Answer: Population growth in a country refers to the increase in its population over time. As per World Bank, India population was around 142 crores in 2023. Cultural and social norms influence population growth in India, affecting family size, birth rates, and fertility patterns.

Social and Cultural norms influence population growth:

- **Institution of Family:** In Indian society, the family is a dominant social institution that influences population growth.
 - For example, parents often pressure their married children to have kids.
- **Son Meta Preference:** The socio-cultural system and patriarchal mindset create a desirable preference for male offspring.
- **Helping Hands:** Population growth is influenced by the notion that more children equate to more hands to earn.
- **Resistance to Contraception:** Religious and traditional beliefs can lead to resistance against the use of contraceptives.
- **Women's Education:** Low levels of female education correlate with higher fertility rates.

Necessity of a Population Control Law:

- **Resource Management:** Rapid population growth strains natural resources like water, land, and energy. A population control law helps manage and conserve these resources for future generations.
- **Poverty Alleviation:** Slower population growth can lead to more equitable distribution of resources, aiding in poverty reduction and improving socio-economic conditions.
- **Addressing Population Stabilization Targets:** As per UN population report, with the current rate of population growth, India is set to miss the population stabilization target.
- **Preparing for an Aging Population:** A progressive population policy can address future issues related to aging. For example, reports indicate that by 2050 every fifth Indian will be over 65.
- **Human Capital Development:** An evidence-based population policy can go beyond mere population control to create a healthy, skilled, and productive population.

Population Control Law is not a necessity because:

- **Declining Total Fertility Rate (TFR):** The Total Fertility Rate (TFR) of the country shows a downward trend, with the majority of states having a TFR below the replacement level i.e. 2.1
- **Reproductive rights concern:** Restrict reproductive rights is unconstitutional, as upheld by the Supreme Court in the Suchita Srivastava case under Article 21.
 - For example, the Uttar Pradesh Law Commission's draft of the Population Control Bill 2021 restricts several benefits to encourage a "two-child policy."
- **Gender Imbalance:** It could result in undesirable consequences such as sex-selective abortions and forced sterilizations. It may lead to demographic distortions, as seen in China.

Implement a comprehensive and humane population control law, focusing on education, healthcare, and economic opportunities to ensure sustainable development and improved quality of life.

Q12: "Despite concerted efforts to eliminate it, the caste system has evolved and persisted." Discuss and highlight the factors that have contributed to the continuance of caste-based discrimination in society.
(15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with definition of caste.

Body: Address the demand by highlighting – First mentioning the efforts to eliminate the caste system. Then list ways in which the caste system persists. Lastly, list factors that contribute to continuance of caste-based discrimination.

Conclusion: Conclude with summarizing the topic with way forward or on a positive note.

Answer: Caste refers to a hereditary social division in Indian society, determining a person's occupation, birth and social norms. The caste system has persisted in Indian society for centuries, despite efforts to eradicate its harmful effects through various means.

Efforts to eliminate the caste system:

- **Constitutional Safeguards:** Provisions like the Right to Equality (Article 14), protection against discrimination (Article 15), and the prohibition of untouchability (Article 17) are enshrined in the Constitution.
- **Legal Framework:** Acts such as the Prevention of Atrocities Act, Protection of Civil Rights Act, and laws addressing manual scavenging and rehabilitation, etc aim to combat caste-based discrimination.
- **Reservation Policies:** Reservation in educational institutions, government jobs, and legislative bodies for SCs and STs to promote social and economic upliftment.
- **Political Initiatives:** Dr. B.R. Ambedkar advocated for the annihilation of caste, and affirmative-action policies have been implemented.

Factors that have contributed to the continuance of caste-based discrimination:

- **Social Discrimination:** Lower-caste individuals face discrimination in rights, education, employment, access to public services, etc.
- **Caste-based Politics:** Political parties use caste affiliations for electoral gains, forming alliances and coalitions that reinforce divisions among communities.
- **Marriage Norms:** Caste influences marriage choices, with inter-caste marriages often facing societal opposition and even violence.
- **Cultural Beliefs:** Beliefs around purity and pollution perpetuate caste roles and stereotypes, affecting social interactions and employment opportunities.
- **Implementation Gaps:** Weak implementation and enforcement of anti-discrimination laws and policies.

Way Forward:

- **Inclusive Curriculum:** Integrate caste-awareness and anti-discrimination education into school curriculums to foster inclusive attitudes from a young age.
- **Strict Enforcement:** Ensure the strict enforcement of existing anti-discrimination laws and policies.
- **Public Awareness Initiatives:** Launch nationwide campaigns to educate and sensitize people about the harms of caste discrimination.
- **Empowerment Initiatives:** Support NGOs and grassroots organizations working towards the empowerment of lower caste communities.

The eradication of the caste system requires continuous efforts across various sectors to address the root causes of discrimination, foster social integration, and empower marginalized communities.

Q13: The foundation of social harmony and national unity is built on 'working with difference', rather than 'working to eliminate differences.' Comment (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introduction about India's core values i.e. social harmony and national unity.

Body: Address the demand by highlighting – what leads to when we work with differences and also mention what happens when we tend to work to eliminate difference. Also, mention way forward.

Conclusion: Conclude with summarizing this on a positive note.

Answer: India is celebrated for valuing diversity, which enhances social harmony and national unity by respecting and celebrating differences, creating a lively and inclusive society.

Working with differences leads to:

- **Cultural Enrichment:** Create a rich, and culturally diverse society by preserving different languages, ethnicities, and religions.
- **Social Cohesion:** A diverse society is more resilient to divisive forces by fostering mutual understanding, respect, and solidarity among different cultural groups.
- **Salad Bowl Model:** Promotes a "salad bowl" model where everyone retains their unique identity while being part of a larger culture.
 - For example, celebrating both classical and tribal dances.
- **Inclusive Policies:** Prevents the alienation of minorities and strengthens national unity.
 - For example, initiatives like Nai Roshni and Jiyo Parsi.

Working to eliminating differences leads to:

- **Loss of Identity:** A push towards uniformity leads to cultural barrenness and defeats the idea of India.
- **Polarization:** Imposing uniformity can deepen divides, creating an "us vs. them" mentality.
 - For example, alienation can push individuals towards radicalization.
- **Resentment and Social Unrest:** It creates a situation where differences are seen as conflicting rather than enriching, leading to tension and division within society.
- **Authoritarianism:** Efforts to eliminate differences often lead to authoritarian policies and governance, undermining democratic principles.
- **Undermines constitutional values:** that protect cultural and religious diversity.

Way Forward:

- **Promote mutual trust and harmony:** For example, inter-faith dialogues and inter-faith marriages.
- **Implement reforms progressively** rather than disruptively. For example, prioritize codification and internal reforms first then proceed with a uniform civil code.
- **Constantly monitor** and control hate speech in social, print, and electronic media to maintain harmony.
- **Encourage moral leadership** from civil society, religious leaders, and politicians to preserve unity.
- **Grievance redressal:** Amicable settlements of inter-state disputes. For example, the Namsai Declaration between Assam and Arunachal Pradesh.

Promoting education that emphasizes the value of diversity, implementing policies that ensure equitable representation, and encouraging community interactions can further enhance social harmony and national unity.

Q14: "A growing sense of regionalism may threaten India's unity and integrity". Critically Examine.

(15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introduction about regionalism.

Body: Address the demand by highlighting – How Regionalism Threatens India's Unity and Integrity and then mention the positives of regionalism.

Conclusion: Conclude with summarizing this on a positive note.

Answer: Regionalism is a political ideology that focuses on the interests of a particular region, group of regions, or other subnational entities. In India, regionalism has posed threats to the country's unity and integrity in several ways.

How Regionalism Threatens India's Unity and Integrity:

- **Disrupting National Identity:** Regionalism making it difficult for people to identify with a unified Indian identity.
 - For example, people often identify themselves based on their states, such as Tamilian, Bengali, Bihari, etc., more than as 'Indian'.
- **Demands for Separatism:** Regionalism can escalate into demands for separatism, threatening India's unity.
 - For example, demands for a separate Khalistan state.
- **Ethnic Tensions:** Regionalism can cause tensions between different ethnic groups, leading to conflicts and violence.
 - For example, ethnic tensions in Northeast India.
- **Economic Disparities:** Regionalism can create economic disparities between regions, leading to feelings of neglect and resentment among people in less developed areas.
 - For example, Marathwada and Vidarbha region.

Benefits of Regionalism:

- **Strengthening Democracy:** Regionalism leads to the rise of regional parties, which tend to address regional socio-economic development issues.
 - For example, the rise of regional parties like DMK, BSP, AAP, and Trinamool Congress.
- **Economic Development:** Regionalism can promote economic development by encouraging investment in less developed regions.
 - For example, the creation of separate states like Chhattisgarh, Telangana, etc has led to their economic development.
- **Improved Governance:** Regionalism can encourage the devolution of power from the central government to local authorities, leading to more efficient and effective governance.
 - For example, effective implementation of schemes like the Mid-day Meal Scheme in Tamil Nadu.
- **Preservation of Local Culture:** Regionalism can help preserve local cultures and traditions, promoting the use of local languages and dialects.
 - For example, Tamil Nadu's regional parties work to secure Tamil language.

Balancing regional aspirations with the broader goals of national integration through effective governance, inclusive policies, and sustained efforts toward national integration are key in managing the potential negative consequences of regionalism.

Q15: Highlight the factors that lead to 'internal migration' within the country and assess its diverse socio-economic effects. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with data on internal migration or with the definition of internal migration.

Body: Address the demand by highlighting – various factors behind the internal migration. Next, analyse the socio-economic impacts of internal migration.

Conclusion: Conclude with way forward to address stressed migration.

Answer: Migration can be defined as the movement of people from one place to another. According to the 2011 Census, there are approximately 450 million internal migrants in India.

Factors influencing internal migration:

- **Economic Opportunities:** Migration often occurs in search of better job opportunities, higher wages, or economic stability
- **Education and Skills:** Access to education and skill development can influence migration patterns.
- **Better Amenities:** Development of cities and infrastructure can attract migrants looking for better living conditions, healthcare facilities, and amenities.
- **Social Factors:** Factors such as marriage, family reunification, caste anonymity can also drive internal migration.
- **Environmental Factors:** Natural disasters, climate change, and environmental degradation can force people to migrate from affected areas to more secure locations.

Positive Socio-Economic Impacts:

- **Composite culture:** Creation of a composite culture, example Chhath Puja being celebrated in Mumbai.
- **Social Mobility:** Migrants often achieve higher social mobility, improving their socio-economic status and quality of life.
- **Liberal Values:** Circular migration helps break stereotypes by introducing modern social values such as equality, family planning, rationality, objectivity, and women's empowerment.
- **Economic growth:** Migrants provide essential labour for industries and services, boosting economic productivity.

Negative Socio-Economic Impacts:

- **Urban slums:** Development of urban slums leads to messy and hidden urbanization, such as Dharavi slum in Mumbai and Bhalswa slum in Delhi.
- **Income Disparities:** Migration can exacerbate income disparities between urban and rural areas, as well as within urban areas.
- **Rise in crime:** There is a rise in urban crimes, including drug smuggling, child trafficking, gender crimes, etc.
- **Social Tensions:** Influx of migrants can lead to social tensions and conflicts over resources, jobs, and cultural differences.
- **Mental health issues:** arise due to individualism and the breakdown of joint families.
- **Negative Impact on rural areas:**
 - Ghost villages due to massive outmigration.
 - Erosion of native culture.

Promote balanced regional development, enhance rural infrastructure, and create sustainable livelihood opportunities to manage internal migration effectively.

Q16: Explain the reasons of global variations in ocean water density. Also, highlight the broader implications of it with the help of examples. (15 marks, 250 words)

Approach to the answer:

Introduction: Give a brief background about ocean water density.

Body: Address the demand by highlighting the difference in density of ocean waters across the globe followed by account for the difference in density of ocean waters. Then, highlight the diverse implications of the difference in density of ocean waters.

Conclusion: Conclude by emphasizing the wide-ranging impacts of density variations.

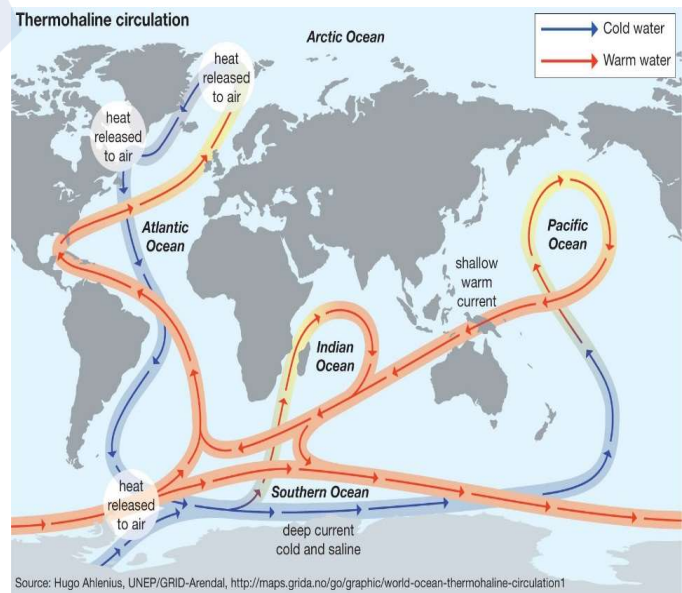
Answer: Ocean water density is the mass per unit volume of seawater, typically expressed in kilograms per cubic meter (kg/m^3). It is primarily influenced by three factors i.e. temperature, salinity and pressure. These variations in density drive ocean circulation patterns.

Factors affecting the ocean water density:

- **Temperature Differences:** Warm water is less dense than cold water, affecting density variations.
 - Example: Tropical seas are warmer and less dense compared to polar seas.
- **Salinity Effect:** Higher salt content increases water density, impacting oceanic circulation patterns.
 - Example: The Atlantic Ocean's higher salinity makes its water denser than the Pacific.
- **Pressure Effects:** Deep ocean layers experience higher pressure, increasing water density.
 - Example: Deep-sea trenches like the Mariana Trench have denser water due to extreme pressures.
- **Seasonal Changes:** Temperature and freshwater input vary seasonally, altering local density.
 - Example: Monsoon rains bring fresh water which reduce salinity in the Bay of Bengal, affecting its water density.
- **Geographical Factors:** Coastal upwelling and currents influence local density gradients.
 - Example: The Red Sea and Mediterranean stay salty because they don't get fresh water from river or melting glaciers.
- **Ocean Currents:** Circulation patterns redistribute water with varying densities across the globe.
 - Example: The Gulf Stream transports warm, less dense water to the North Atlantic.

Implications:

- **Thermohaline Circulation:** Density differences due to temperature and salinity drive the global conveyor belt, regulating heat and nutrient distribution.
 - Example: The thermohaline circulation helps maintain moderate temperatures in Western Europe through the Gulf Stream.
- **Ecological Marine Ecosystems:** Density gradients affect nutrient transport, impacting marine biodiversity and fisheries.
 - Example: Upwelling zones support rich fisheries due to nutrient-rich denser water rising to the surface.
- **Economic Significance:** Saltwater's lower freezing point helps ships move through icy waters, which supports global shipping and fishing industries.
- **Energy Potential:** Differences in temperature between the ocean's surface and deeper layers can be used for Ocean Thermal Energy Conversion (OTEC), a way to generate renewable energy.
- **Climate Regulation:** Ocean density influences the storage and release of heat. Dense and cold waters can store more carbon dioxide, affecting atmospheric CO_2 levels and global temperatures.



Understanding these variations is crucial for predicting climate change impacts, managing marine resources, and protecting coastal communities.

Q17: Examine the role of the Indian Ocean Dipole and El Niño-Southern Oscillation (ENSO) in influencing the frequency and intensity of tropical cyclones in the Indian Ocean region. (15 marks, 250 words)

Approach to the answer:

Introduction: Give a brief definition of IOD and ENSO.

Body: Address the demand by highlighting IOD and ENSO influence on tropical cyclone.

Conclusion: Conclude with either summarizing the topic or with climate change further impacting tropical cyclones.

Answer: The Indian Ocean Dipole (IOD) refers to the alternating temperature gradient and associated atmospheric circulation changes in the Indian Ocean, while El Niño-Southern Oscillation (ENSO) involves fluctuations in sea surface temperatures across the central and eastern Pacific Ocean.

Understanding the complex interplay of these two is crucial for predicting and managing the impacts of tropical cyclones in the Indian Ocean region.

- **IOD Influence on Cyclone Frequency and Intensity:**
 - **Positive IOD Phase:** Increases Sea surface temperatures in the western Indian Ocean enhance atmospheric instability and increase moisture, promoting cyclone formation and intensification.
 - **Negative IOD Phase:** Decreases Sea surface temperatures in the western Indian Ocean, suppresses cyclone activity due to reduced atmospheric convection.
- **ENSO Impact on Cyclone Patterns:**
 - **El Niño conditions:** Warmer waters in the central and eastern Pacific, lead to increased wind gradient over the Indian Ocean. This inhibits cyclone development and intensification.
 - **La Niña:** Cooler waters in the central and eastern Pacific, reduces wind gradient, allowing for more favorable conditions for cyclone formation and stronger storms in the Indian Ocean.
- **Combined Influence:**
 - The interaction between IOD and ENSO can amplify or reduce their individual effects on cyclone activity.
 - For example, a positive IOD coupled with La Niña can lead to active cyclone seasons in the Indian Ocean.

Understanding these phenomena is essential for predicting cyclone behavior and improving disaster preparedness. With climate change altering ocean temperatures and atmospheric dynamics, monitoring and adapting to these influences will become increasingly important in mitigating cyclone-related risks.

Q18: Highlight various landforms of karst region. Also, discuss the ecological and hydrological significance of karst regions. (15 Marks, 250 Words)

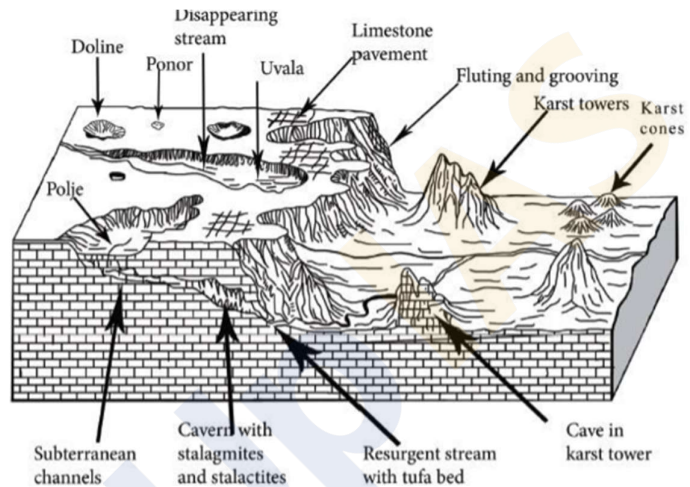
Approach to the answer:

Introduction: Give a brief background of Karst region.

Body: Address the demand by highlighting the various landforms formed in karst region and also mention ecological and hydrological significance of karst regions.

Conclusion: Conclude with conservation efforts on a positive note.

Answer: A karst region is a landscape formed from the dissolution of soluble rocks such as limestone, dolomite, and gypsum. This process creates distinctive landforms and unique underground drainage systems. Karst topography is found in many regions around the world (Yugoslavia on the Adriatic coast) and has significant ecological, geological, and human importance.



Various Landforms of Karst Region

- **Sinkholes (Dolines):** Depressions or holes in the ground caused by the collapse of a cave roof or the dissolution of surface limestone.
- **Caves:** Natural underground spaces, often formed by the dissolution of limestone, featuring stalactites and stalagmites.
- **Poljes:** Large, flat-floored depressions in karst areas, typically with a seasonal or permanent water body.
- **Limestone Pavements:** Exposed flat surfaces of limestone that have been weathered to create blocks (clints) separated by fissures (grikes).
- **Uvalas:** Large closed depressions formed by the merging of several sinkholes.
- **Disappearing Streams:** Surface streams that disappear into the ground, often feeding into an underground cave system.
- **Karst Springs:** Points where groundwater flows out of the ground, often at the base of limestone cliffs or in valley bottoms.

Ecological and Hydrological Significance:

- **Biodiversity Hotspots:** Karst landscapes support diverse ecosystems adapted to cave and limestone habitats.
- **Critical Habitats:** Caves and sinkholes provide habitats for specialized species. Example: Blind cavefish in Mexican cenotes.
- **Water Storage and Supply:** Karst regions act as natural reservoirs, storing and slowly releasing groundwater essential for streamflow and human consumption.
- **Natural Filtration:** Karst springs provide clean, mineral-rich water crucial for freshwater ecosystems.

Karst systems, sensitive to surface pollutants due to rapid water movement through porous rocks, require sustainable management efforts and community efforts to protect ecosystems, manage water resources for long-term sustainability.

Q19: Explain the geological processes that led to the formation of the Himalayan Mountain range. Also, analyse the role of the Himalayas in shaping the monsoon patterns in the Indian subcontinent.
(15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with some factual information about Himalaya.

Body: Address the demand by highlighting Geological Processes leading to formation and role of Himalaya's in shaping monsoon patterns.

Conclusion: Conclude with summarizing the topic.

Answer: The Himalayan Mountain range, spanning over 2,400 kilometers across five countries i.e. India, Nepal, Bhutan, China, and Pakistan, is the youngest and highest mountain range in the world. The geological evolution of the Himalayas can be attributed to the movement and interaction of Earth's tectonic plates over millions of years.

Geological Processes Leading to Formation:

- **Karakoram Phase:** The Indian and Eurasian plates began colliding. This collision compressed sedimentary rocks, creating folds and thrusts.
- **Malla Johar Phase:** The Himalayan range started forming as the Indian plate moved northwards, causing the Tethys Sea's oceanic crust to subduct under the Eurasian plate. This compression and folding of rocks formed the Great Himalayan range.
- **Sirmurian Phase:** The continued northward movement of the Indian plate led to the formation of the Lesser Himalayas.
- **Siwalik Phase:** The ongoing collision between the Indian and Eurasian plates caused further uplift and erosion, forming the Siwalik range from the sedimentary rocks of this period.

Role in Shaping Monsoon Patterns

- **Orographic Effect:** The Himalayas act as a barrier to the monsoon winds, forcing them to rise and cool, leading to heavy rainfall in the Indian subcontinent.
- **Rain Shadow Effect:** The mountains block the monsoon winds, causing dry conditions in the regions lying to their north. E.g. Great Indian Cold Desert.
- **Temperature Regulation:** The Himalayas influence the temperature gradients, which in turn affect the intensity and onset of the monsoons.
- **Jet Streams Interaction:** The presence of the Himalayas alters the path of the jet streams, impacting the timing and distribution of the monsoon rains.

The collision of the Indian and Eurasian tectonic plates led to the uplift and ongoing growth of the Himalayan Mountains. These mountains significantly influence the life-sustaining monsoons of the Indian subcontinent, demonstrating the complex interaction between geological processes and climatic patterns.

Q20: Discuss the reasons behind the increasing incidents of urban floods in Indian cities. Also, suggest measures to mitigate them.
(15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with phenomena of Urban Flood in brief.

Body: Address the demand by main causes of urban flooding in India then mention measures to address this.

Conclusion: Conclude with summarizing the topic or way forward.

Answer: Urban flooding occurs due to excessive runoff in developed urban areas, where stormwater overwhelms drainage systems, causing inundations. Heavy rainfall linked to climate change and rapid urbanization are primary causes.

Main Causes of Urban Flooding in India:

- **Meteorological Factors:** Increased heavy rainfall during monsoons and extreme weather events due to climate change result in higher water runoff.
 - Urban Heat Island effect alters precipitation patterns.

- **Hydrological Factors:** Rapid urbanization disrupts natural drainage patterns, increases surface runoff, and reduces percolation due to concretization, deforestation, and soil erosion.
- **Anthropogenic Factors:**
 - **Neglect of Natural Topography:** Poor consideration of natural terrain inhibits proper water drainage during heavy rainfall. Ineffective protection of water bodies contributes to recurrent flooding.
 - **Urbanization:** Impervious surfaces restrict water absorption, heightening flood risks.
 - **Encroachment:** Settlements in low-lying areas impede natural water flow, exacerbating flooding.
 - **Clogged Drainage Channels:** Improper waste disposal and silted channels hinder water drainage.

Mitigation Strategies for Urban Flooding:

- **Monitoring and Preparedness:** Analyzing topography and historical flood data to identify vulnerable areas and enhance readiness.
- **Water Management:** Regular maintenance of drainage channels, preservation of catchment areas, and rainwater harvesting to prevent waterlogging and pollution.
- **Urban River Management:** Balancing development with conservation and restoration of urban rivers for ecological and economic benefits.
- **Sponge City Concept:** Implementing urban design strategies to enhance rainwater absorption and storage, reducing flood risks and improving water quality.
- **Infrastructure Upgradation:** Building new stormwater drains and expanding sewerage facilities to improve urban infrastructure resilience.

While the government initiatives are crucial, community engagement and public education are essential for effective disaster mitigation and management in urban areas prone to flooding.